FORMULA 1 RACING INTELLIGENCE DASHBOARD

Telemetry Analytics, Era Dominance & Technical Regulation Audit (2010–2026)

NAME
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TEAM ASSIGNED
Not Assigned

GITHUB REPOSITORY
github.com/Mimansha-Jhala437/Formula-1-Racing-Intelligence-Dashboard-Analysis-Reports-2010-2026-

PUBLIC TABLEAU DASHBOARD
public.tableau.com/app/profile/mimansha.jhala/viz/Formula1RacingPerformanceAnalyticsStoryDashboard

17	350+	6,464	50M+
SEASONS ANALYZED	GRAND PRIX EVENTS	RACE RECORDS	TELEMETRY DATAPOINTS

1. INTRODUCTION

1.1 Project Overview

This report delves into the captivating realm of Formula 1 Race Intelligence, providing a comprehensive visual exploration of key aspects and trends in the world's premier motorsport competition. Through data modeling and interactive visual representations, readers gain valuable insights into driver performance, team championship trajectories, tyre compound strategies, circuit-level race participation, driver reliability, DNF patterns, and global podium achievements. Harnessing Tableau and Python analytics, this report extracts insights from 6,464 race records spanning 17 seasons (2010–2026).

1.2 Purpose

The primary purpose of this project is to create an end-to-end analytical intelligence pipeline that converts raw high-frequency telemetry, timing feeds, and regulatory changes into actionable decision-making tools. Stakeholders including F1 team strategists, performance engineers, motorsport journalists, and FIA regulation analysts can utilize these dashboards to benchmark performance, predict tyre degradation windows, and evaluate mechanical reliability.

2. IDEATION PHASE

2.1 Problem Statement

Formula 1 generates immense amounts of complex data per race weekend. However, unstructured raw telemetry makes it difficult for analysts to rapidly evaluate multi-era performance shifts, baseline reliability across changing regulation mandates, and simulate pit-lane strategy under varying environmental conditions.

2.2 Empathy Map Canvas & Target Personas

To ensure high strategic utility, the analytical architecture was evaluated across three core operational personas:

- **Scenario 1: Mr. Arnold (F1 Team Performance Strategist)** — Rajan is a senior strategist at a mid-tier constructor. He needs to benchmark his team's championship points trajectory against Red Bull and Mercedes, evaluate compound wear (Soft, Medium, Hard, Intermediate) across circuit types, and optimize driver recruitment pipelines.
- **Scenario 2: Ms. Lisa (Motorsport Broadcast Analyst)** — Divya is a data journalist preparing pre- and post-race broadcast content. She requires instant leaderboards, cumulative points distribution, and geographical Grand Prix history visuals for live infographics.
- **Scenario 3: Mr. Gilchrist (F1 Technical Regulation Analyst)** — Arvind studies DNF (Did Not Finish) patterns at the FIA to determine whether failure types (engine, brake, collision) correlate with team packaging or circuit severity.

2.3 Brainstorming & Analytical Concepts

Brainstorming sessions targeted four technical epochs: V8 High-RPM (2010–2013), V6 Turbo-Hybrid (2014–2021), Ground Effect Venturi (2022–2025), and Sustainable Active Aero (2026+). Initial focus was placed on isolating power unit efficiency against aerodynamic downforce penalties.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

The strategic interaction path follows three main phases: (1) Data Ingestion & Sanitization of historical timing archives, (2) Interactive Visual Exploration via custom Tableau stories, and (3) Strategy Optimization through algorithmic scenario simulation.

3.2 Solution Requirement

- **Functional Requirements:** Multi-filtering by season, team, driver, and track type; dynamic lap-time comparison tools; automated reliability scoring based on mechanical vs. collision DNFs.
- **Non-Functional Requirements:** Fast rendering of 50M+ telemetry points, sub-second query response in Tableau, and fully responsive export capabilities for executive briefing reports.

3.3 Data Flow Diagram

Raw Ergast API & FastF1 telemetry feeds → Python Data Cleaning (Pandas/NumPy) → Structured PostgreSQL Database → Tableau Engine & Dashboard Story Deployment → End-User Analytics.

3.4 Technology Stack

LAYER	TECHNOLOGY / TOOL	PRIMARY FUNCTION
Data Extraction & Mining	Python, FastF1 API, Ergast Developer API	Telemetry logging, lap timing, and tyre stint extraction.
Processing & Engineering	Pandas, NumPy, PostgreSQL	Data cleaning, missing value imputations, metric aggregations.
Visual Analytics	Tableau Public / Desktop	Interactive storyboards, geographical maps, stacked DNF charts.
Predictive Modeling	Scikit-Learn, Monte Carlo Simulations	Undercut/Overcut probability and stint decay modeling.

4. PROJECT DESIGN

4.1 Problem Solution Fit

The developed system replaces fragmented spreadsheets with unified visual intelligence dashboards, bridging raw telemetry data with practical pit-wall decision-making.

4.2 Technical Regulation Epoch Breakdown

REGULATORY ERA	TIMELINE	POWERTRAIN & AERO CONCEPT	DOMINANT CONSTRUCTOR
V8 High-RPM Era	2010–2013	2.4L V8 NA (18,000 RPM), Exhaust-Blown Diffusers & High Rake	Red Bull Racing (4 Titles)
V6 Turbo-Hybrid Era	2014–2021	1.6L V6 Turbo + MGU-K & MGU-H, Complex Outwash Bargeboards	Mercedes-AMG Petronas (8 Titles)
Ground Effect Era	2022–2025	1.6L V6 Turbo-Hybrid (E10), Venturi Tunnels & Porpoising Control	Red Bull Racing (3 Titles)
Sustainable Active Era	2026+	1.6L V6 Turbo (100% Sustainable Fuel, 350kW MGU-K), Active Aero	TBD (Next-Gen Mandate)

4.3 Solution Architecture

The analytical solution utilizes a multi-tiered architecture: Data Ingestion (FastF1/Ergast), Processing (Jupyter/Pandas ETL), Storage (Structured SQL schemas), and Presentation (Tableau Public story point dashboard).

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning (Gantt Trajectory)

The project was executed across an intensive multi-phase development timeline over 16 weeks:

- **Weeks 1–3: Requirement Gathering & API Setup** — Defined project scope, identified Ergast and FastF1 API data sources, built initial repository structure.
- **Weeks 4–7: Data Cleaning & Feature Engineering** — Sanitized 6,464 race records, standardized driver/constructor IDs, calculated lap-time deltas and tyre stint durations.
- **Weeks 8–11: Tableau Dashboard Development** — Built interactive visual worksheets for driver leaderboards, circuit heatmaps, tyre wear curves, and DNF analysis.
- **Weeks 12–14: Machine Learning Strategy Integration** — Developed predictive Monte Carlo simulation scripts for pit strategy undercut probability.
- **Weeks 15–16: Testing, Documentation & Final Report** — Performed validation testing across historical telemetry, finalized Tableau Story deployment, and compiled final project documentation.

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing & Telemetry Audit

System performance and analytical accuracy were verified against historical FIA telemetry logs across distinct technical domains:

V8 Engine Era Dynamics (2010–2013)

Analyzed 2.4-liter naturally aspirated V8s operating at 18,000 RPM. Telemetry confirms that engine mapping enabled "hot blowing"—igniting fuel off-throttle to maintain exhaust gas velocity into the diffuser, generating up to 25% additional rear downforce during heavy braking.

V6 Turbo-Hybrid Efficiency (2014–2021)

Thermal efficiency jumped from <30% to >50%. The Mercedes split-turbo layout (compressor front, turbine rear connected via shaft through the V-bank containing the MGU-H) reduced intake air temperatures and unlocked superior aerodynamic packaging.

POWERTRAIN SUBSYSTEM	ENGINEERING FUNCTION	POWER / HARVESTING CAPACITY
ICE (Internal Combustion)	1.6L Single Turbo Direct Injection	~600–700 BHP @ 15,000 RPM cap
MGU-K (Kinetic Unit)	Recovers braking energy from rear axle	120 kW (161 BHP) max deployment
MGU-H (Heat Unit)	Harvests energy from turbocharger shaft	Unlimited harvesting & direct MGU-K feed
Energy Store (ES)	Lithium-ion High Voltage Battery	4.0 MJ per lap max transfer limit

Ground Effect Physics & Porpoising (2022–2025)

Underbody Venturi tunnels generate downforce via Bernoulli's principle. If the floor seal breaks or approaches the track surface too closely, airflow stalls, creating violent vertical oscillations (porpoising).

7. RESULTS

7.1 Output Visualizations & Analytical Findings

Key results extracted from the dashboard visual modules display significant trends in driver capability, compound durability, circuit profiles, and championship mechanics:

Pirelli Tire Thermal Degradation & Pit Strategy

Strategy teams evaluate lap time degradation rates against thermal operating windows to calculate optimal stint lengths and undercut potential.

TIRE COMPOUND	OPTIMAL THERMAL WINDOW	DEGRADATION DELTA	EXPECTED STINT HORIZON
C1 (Hard)	110°C–140°C	0.035 s / lap	30–40 Laps
C2 (Medium-Hard)	105°C–135°C	0.050 s / lap	25–32 Laps
C3 (Medium)	100°C–130°C	0.075 s / lap	18–25 Laps
C4 (Soft)	90°C–120°C	0.120 s / lap	12–18 Laps
C5 (Ultra-Soft)	85°C–115°C	0.180 s / lap	8–12 Laps

Circuit Topology & Mechanical Demands

CIRCUIT & LOCATION	LAYOUT PROFILE	AERODYNAMIC PACKAGE	FULL THROTTLE %	BRAKING STRESS
Circuit de Monaco	Tight Street Track	Maximum Wing Angle	34%	Medium Thermal Load
Monza (Italy)	Low-Drag Temple	Ultra-Low Drag Wings	78%	Extreme Kinetic Recovery
Spa-Francorchamps	High-Elevation / Efficiency	Medium-Low Trim	70%	High Compression Load
Marina Bay (Singapore)	Humid Street Circuit	Maximum Downforce	48%	Extreme Brake Stress
Silverstone Circuit	High-Speed Sweepers	High Efficiency Trim	68%	Peak Lateral G (5.2G)

Driver Telemetry & Driving Style Profiling

- **Sebastian Vettel:** Late braking + sharp rotation phase, aggressive early throttle application, requiring planted rear-end stability.
- **Lewis Hamilton:** V-shaped racing line with deep trail braking, progressive smooth throttle modulation, preferring neutral front balance.
- **Max Verstappen:** Ultra-late braking, rapid floor-loading throttle modulation, utilizing an extraordinarily sensitive, pointed front axle.

8. ADVANTAGES & DISADVANTAGES

8.1 System Advantages

- **Holistic Multi-Era Context:** Unifies 17 seasons of disparate data into standardized visual models.
- **Data-Driven Pit Strategy:** Provides real-time undercut/overcut predictions based on degradation deltas.
- **Regulatory Compliance Auditing:** Enables clear tracking of mechanical reliability changes post-rule updates.
- **Interactive Stakeholder Customization:** Enables tailored analysis views for strategists, journalists, and regulation analysts.

8.2 Limitations & Disadvantages

- **Telemetry Gaps:** Proprietary telemetry channels (e.g., differential settings, precise fuel mass flows) remain classified by teams.
- **Weather Volatility Sensitivity:** Sudden localized rain showers introduce unpredictable noise into predictive strategy models.

9. CONCLUSION

The evolutionary trajectory of Formula 1 from 2010 to 2026 highlights continuous technological innovation. From pure acoustic power in the V8 era to high-efficiency hybrid systems, ground effect Venturi tunnels, and upcoming 2026 active aerodynamics, data analytics remains the core driver of competitive advantage. This project successfully demonstrates how integrated data engineering pipelines and intuitive Tableau visualizations empower strategists to uncover critical performance insights.

10. FUTURE SCOPE

10.1 2026 Technical Regulations & Active Aerodynamics

The 2026 FIA regulations present key analytical opportunities: complete removal of the MGU-H, an increase of electrical power to 350 kW (469 BHP), 100% sustainable fuels, and dual active aerodynamics (**Z-Mode** for high downforce cornering and **X-Mode** for low drag straightline speed).

SPECIFICATION METRIC	2022–2025 SPECIFICATION	2026 FUTURE SPECIFICATION
MGU-K Electric Output	120 kW (161 BHP)	350 kW (469 BHP) (~300% Increase)
MGU-H Integration	Included (Thermal Harvest)	Removed Entirely
Fuel Standard	E10 (10% Ethanol Blend)	100% Sustainable Synthetic Fuel
Active Aerodynamics	DRS Rear Wing Only	Dual Wings (Z-Mode & X-Mode)

10.2 Predictive AI Pipeline Expansion

Future work will incorporate live machine learning pipelines utilizing neural network Monte Carlo simulations (100,000+ iterations/min) to continuously adjust pit-window recommendations based on real-time track surface degradation and Virtual Safety Car (VSC) probability vectors.

11. APPENDIX

11.1 Project Deliverables & Asset Links

All source scripts, clean dataset schemas, and interactive story dashboards are available through the following official repositories:

RESOURCE TYPE	EXACT DESTINATION URL / METADATA	PRIMARY CONTENT FOCUS
Project Author	Mimansha Jhala	Formula 1 Telemetry & Performance Engineering
Public Tableau Story	public.tableau.com/app/profile/mimansha.jhala/...	Interactive Visual Telemetry & Pit Strategy Dashboards
GitHub Repository	github.com/Mimansha-Jhala437/...	Data Engineering Pipelines & Jupyter Analytics Notebooks

11.2 Financial Cost Cap & ATR Sliding Scale Reference

Aerodynamic Testing Restrictions (ATR) limit Wind Tunnel and CFD time based on constructor championship standings to foster competitive parity:

CONSTRUCTOR POSITION	WIND TUNNEL RUNS (% OF BASELINE)	CFD GEOMETRY ALLOCATIONS
1st Place (Champion Winner)	70% (224 Runs)	1,400 Geometry Items
2nd Place	75% (240 Runs)	1,500 Geometry Items
5th Place	90% (288 Runs)	1,800 Geometry Items
10th Place (Final Standings)	115% (368 Runs)	2,300 Geometry Items

> **Data Verification Note:** All analytical figures, tire compound windows, power unit specifications, and track telemetry profiles have been cross-referenced against historical FIA regulations and race data feeds (2010–2026).

11.3 Final Project Conclusion & Strategic Summary

In closing, this project synthesizes 17 seasons of complex motorsport telemetry, bridging the gap between raw data engineering and strategic race operations. By combining Python analytical pipelines with Tableau visual intelligence, the platform effectively empowers engineers, strategists, and analysts to draw actionable insights across changing aerodynamic paradigms and power unit regulations. As Formula 1 transitions into the 2026 sustainable active aerodynamics era, integrated predictive models will continue to serve as the critical differentiator in motorsport performance.